

ML-Based Dengue Outbreak Prediction System for the Colombo District

R26-IT-061

Project ID: R26-IT-061

Project Proposal Report

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ABSTRACT

This research focuses on developing a machine learning-based dengue outbreak prediction system using historical dengue case data and weather variables such as rainfall, temperature, humidity, and wind speed. Dengue remains a major public health problem in tropical regions including Sri Lanka, where outbreaks occur frequently and place pressure on healthcare systems. Current dengue monitoring methods rely mainly on retrospective statistical analysis, which limits the ability to detect outbreaks early and respond proactively.

The identified research gap is the lack of localized, explainable, and integrated predictive systems that combine machine learning models with climate data and intelligent knowledge interpretation. Many existing studies focus only on national-level predictions and operate in offline experimental environments without providing interpretable outputs for decision makers.

The proposed solution is an integrated dengue prediction platform that combines machine learning algorithms, explainable analytics, and natural language processing. Machine learning models such as Random Forest, XGBoost, and LSTM will be trained to identify complex relationships between weather conditions and dengue outbreaks. The system will generate weekly risk predictions categorized as low, medium, or high. Additionally, a knowledge-augmented NLP component will retrieve dengue-related information from curated data sources and generate evidence-based insights using retrieval-augmented generation techniques.

Technologies used include Python, machine learning libraries, vector databases, graph databases, transformer models, and interactive data visualization dashboards. The expected outcome is a decision-support system that provides early warnings and interpretable insights for public health authorities. The project has strong social impact by supporting dengue prevention strategies and improving public health decision making.

Keywords: Dengue Prediction, Machine Learning, Climate Data, NLP, Public Health Intelligence

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1. INTRODUCTION

1.1 Background and Context

Dengue fever is one of the most significant mosquito-borne viral diseases affecting tropical and subtropical countries. Sri Lanka experiences frequent dengue outbreaks, especially in densely populated districts such as Colombo. Climatic factors such as rainfall, temperature, and humidity strongly influence mosquito breeding and disease transmission.

Traditional dengue surveillance systems mainly analyze historical case records using descriptive statistics. Although these approaches provide insights into past trends, they cannot accurately predict future outbreaks. As a result, public health responses often occur after case numbers have already increased.

Machine learning techniques have demonstrated strong potential in modeling complex nonlinear relationships between climatic factors and disease transmission. By integrating historical dengue data with weather variables, predictive models can generate early warnings that enable preventive interventions.

This research aligns with Sustainable Development Goal 3 (Good Health and Well-being) by supporting disease prevention and improving public health decision making.

1.2 Literature Review

Several studies have investigated dengue outbreak prediction using statistical and machine learning techniques. Traditional models such as linear regression and ARIMA have been widely used to analyze dengue trends. However, these models have limitations in capturing nonlinear relationships and complex interactions between environmental variables.

Recent research has explored machine learning algorithms including Random Forest, Support Vector Machines, Gradient Boosting, and Artificial Neural Networks. These models demonstrate improved prediction accuracy when analyzing climate-driven disease patterns. Despite these advancements, many studies focus on national-level data and lack localized prediction capabilities.

Additionally, existing systems often function as black-box models that provide predictions without explaining the underlying factors influencing the results. This lack of transparency reduces trust among public health officials and limits real-world adoption.

1.3 Research Gap

Although previous studies have shown the effectiveness of machine learning for dengue prediction, several research gaps remain. Many systems lack localized prediction models tailored to specific regions. In addition, most approaches do not integrate explainable AI techniques that allow decision makers to understand the reasons behind predictions.

Another major limitation is the absence of intelligent knowledge systems that combine machine

learning predictions with contextual information from health reports, research publications, and news sources. Addressing these limitations requires an integrated framework that combines predictive analytics, explainable models, and knowledge-based NLP systems.

Gap Dimension	Existing Research / Systems	Identified Limitation	Proposed Contribution
Data Engineering as Research Problem	Existing dengue prediction studies (Hii et al., Xu et al., Pham et al.) apply a fixed, single preprocessing pipeline without evaluating alternatives.	Pipeline design decisions are often made arbitrarily, and the impact of different preprocessing strategies on model performance is rarely evaluated systematically.	The research experimentally evaluates multiple preprocessing pipelines including different normalization, cleaning, time-lag generation, and feature selection strategies. Each configuration is treated as a hypothesis and validated through machine learning performance metrics.
Biologically Informed Time-Lag Features	Previous studies that include lag variables (Hii et al., 2012; Tissera et al., 2016) typically use fixed lag periods without biological justification.	Arbitrary lag values may not reflect the real mosquito life cycle or dengue transmission dynamics, reducing the scientific validity of predictions.	This study engineers biologically informed lag features (1–4 weeks) based on the life cycle of <i>Aedes aegypti</i> , the extrinsic incubation period of the dengue virus, and delays in case reporting.
Spatial Resolution Sensitivity Analysis	Prior research such as Nsoesie et al. (2014) highlights the impact of spatial resolution on disease prediction but does not evaluate this within the Sri Lankan context.	There is limited empirical evidence on whether district-level or more granular spatial units provide better predictive accuracy in Sri Lanka.	This research evaluates spatial resolution sensitivity by comparing district-level and MOH area-level datasets and measuring differences in predictive performance using metrics such as F1-score and RMSE.
Multiple ML-Ready Dataset Variants	Existing dengue prediction studies typically rely on a single prepared dataset for model training and evaluation.	Without multiple dataset configurations, researchers cannot assess how different data engineering decisions influence prediction performance.	The study generates multiple ML-ready dataset variants and maintains a structured dataset registry to systematically compare the effect of different data engineering choices on model accuracy.

Gap Dimension	Existing Research / Systems	Identified Limitation	Proposed Contribution
MOH Area-Level Geospatial Linking	Existing dengue surveillance platforms such as WHO monitoring systems and regional dashboards usually operate at national or provincial scales.	These systems do not provide data linked to the Medical Officer of Health (MOH) area, which is the operational level for local public health interventions in Sri Lanka.	The proposed system links dengue case data with meteorological and environmental datasets at the MOH area level within Colombo District using geospatial tools such as GeoPandas and GIS boundary datasets.

2. OBJECTIVES

2.1 Main Objective

To design and develop a machine learning-based dengue outbreak prediction system that utilizes historical dengue case data and meteorological variables to generate accurate and interpretable early warning alerts for dengue risk.

2.2 Specific Objectives

1. Collect and preprocess historical dengue case data and climate datasets.
2. Develop machine learning models to predict dengue outbreak risk levels.
3. Implement explainable AI techniques to interpret model predictions.
4. Design interactive dashboards to visualize dengue risk trends.
5. Develop a knowledge-augmented NLP framework to retrieve and generate dengue-related insights.

3. METHODOLOGY

3.1 Research Approach

This research adopts an Experimental Research methodology within the Design Science Research (DSR) paradigm. The purpose of the study is not only to build a predictive dengue outbreak model but also to systematically evaluate how different data engineering strategies influence machine learning performance.

Instead of proposing a fixed preprocessing pipeline, the key data engineering steps such as data cleaning methods, normalization techniques, lag feature construction, and feature selection approaches are treated as independent experimental variables. Multiple dataset variants will be generated using different combinations of these preprocessing techniques.

The dependent variable in this experiment is the predictive performance of the machine learning models on the dengue outbreak prediction task. Model performance will be evaluated using metrics such as F1-score, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Precision, Recall, and ROC-AUC.

This approach enables empirical evaluation of which data engineering configurations produce the most accurate and reliable dengue outbreak predictions, thereby contributing methodological insights into epidemiological machine learning workflows.

3.2 Data Collection and Ethical Considerations

Two primary sources of data will be utilized for this research component.

National Dengue Control Unit (NDCU)

Weekly dengue case data will be collected for the Colombo District and its Medical Officer of Health (MOH) areas covering the period 2014–2024. This dataset provides more than ten years of epidemiological records, enabling the study to capture multiple dengue outbreak cycles and seasonal patterns.

Department of Meteorology – Sri Lanka

Meteorological variables will be collected on a daily and weekly basis for the same geographic region. The variables include:

- Total rainfall (mm)
- Mean temperature (°C)
- Minimum temperature (°C)
- Maximum temperature (°C)
- Relative humidity (%)

- Wind speed (km/h)
- Sunshine hours

All datasets used in this research consist of aggregated public health and environmental data. No individual patient information will be used. Therefore, personal privacy risks are minimal.

To ensure responsible data usage:

- Data will be anonymized and aggregated
- Data Use Agreements will be established with relevant institutions
- The research proposal will be submitted to the SLIIT Research Ethics Committee (REC) for ethical clearance before the project begins.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis will be conducted using Python and analytical libraries including Pandas, NumPy, Matplotlib, Seaborn, and Statsmodels.

The following analytical steps will be performed:

Time Series Decomposition

Dengue case counts and meteorological variables will be decomposed into trend, seasonal, and residual components to understand long-term patterns and cyclical behavior.

Distribution Analysis

Histograms, box plots, and kernel density plots will be used to analyze the statistical distributions of the variables and identify potential outliers.

Cross-Correlation Analysis

Cross-correlation functions will be used to measure the relationship between climate variables and dengue case counts at lags ranging from 1 to 8 weeks, identifying possible delayed environmental effects.

Missing Data Pattern Analysis

Missing value patterns will be analyzed to determine whether they follow MCAR (Missing Completely at Random), MAR (Missing at Random), or MNAR (Missing Not at Random) distributions.

Geospatial Visualization

Spatial patterns of dengue cases will be visualized using choropleth maps generated with GeoPandas to analyze the distribution of dengue incidence across MOH regions.

3.4 Data Cleaning Strategies

To evaluate the influence of data cleaning approaches, multiple strategies will be implemented and compared.

Strategy A – Mean/Median Imputation

A simple univariate imputation approach where missing values are replaced using the mean or median of the feature. This approach is computationally efficient but may not capture relationships between variables.

Strategy B – Forward Fill (Time-Series Based)

Missing values are replaced using the most recent available observation. This method is particularly suitable for short gaps in time-series health data.

Strategy C – Multiple Imputation by Chained Equations (MICE)

MICE is a multivariate imputation method that estimates missing values iteratively while preserving correlations between variables. This method is especially useful when missing values follow MAR patterns.

Outlier Treatment Methods

Outliers will be treated using three alternative techniques:

- Interquartile Range (IQR) Winsorization using $1.5 \times \text{IQR}$ thresholds
- Z-score filtering using ± 3 standard deviation limits
- Isolation Forest anomaly detection, a machine learning based outlier detection technique

3.5 Normalization and Scaling

Four normalization and scaling methods will be implemented to evaluate their impact on predictive modeling.

Min-Max Scaling

Rescales all features into the $[0,1]$ range. This approach is suitable for rainfall and epidemiological variables with bounded distributions.

Z-score Standardization

Transforms variables so that they have mean = 0 and standard deviation = 1. This method is particularly suitable for normally distributed variables such as temperature.

Robust Scaling

Uses the median and interquartile range (IQR) to scale the data. It is highly resistant to extreme values and is appropriate for datasets containing outbreak spikes.

Log Transformation

A logarithmic transformation will be applied to highly skewed variables such as dengue case counts to stabilize variance and reduce the influence of extreme values.

3.6 Feature Engineering — Biologically Informed Time-Lag Features

The life cycle of the *Aedes aegypti* mosquito, the primary dengue vector, plays a critical role in determining the relationship between environmental conditions and disease outbreaks.

The mosquito typically requires 8–10 days to develop from egg to adult, while the dengue virus undergoes an extrinsic incubation period of 8–12 days within the mosquito. Additional delays may arise due to human incubation periods and reporting delays.

Therefore, environmental conditions may influence reported dengue cases with a delay of approximately 2–4 weeks.

Based on this biological evidence, the following features will be engineered:

Lagged Climate Variables

Climate features will be shifted by:

- 1 week
- 2 weeks
- 3 weeks
- 4 weeks

for rainfall, temperature, humidity, and other meteorological variables.

Rolling Window Features

Moving averages will be calculated over:

- 2-week windows
- 4-week windows

to represent cumulative environmental exposure.

Cumulative Rainfall Indicators

Cumulative rainfall over 2-week and 4-week periods will be calculated to estimate potential mosquito breeding conditions due to stagnant water accumulation.

Each lag configuration will generate distinct dataset variants for experimental comparison.

3.7 Feature Selection

Three complementary feature selection approaches will be applied to identify the most informative predictors.

Filter-Based Methods

Statistical measures will be used to identify relationships between predictors and dengue case counts:

- Pearson correlation
- Spearman rank correlation
- Mutual information scores

Wrapper-Based Methods

Recursive Feature Elimination (RFE) using a Random Forest estimator will iteratively remove less important variables to determine the minimal feature set that preserves predictive performance.

Embedded Methods

Feature importance rankings derived from Random Forest and XGBoost models will also be used to evaluate feature contributions.

The final feature set will be determined using a consensus approach, selecting variables that consistently rank highly across all three methods.

3.8 Geospatial Analysis and Spatial Resolution Sensitivity

Dengue case data will be linked with geographic boundary datasets at two levels of spatial resolution.

District-Level Analysis

All dengue cases within the Colombo District will be aggregated into a single administrative unit. While this approach provides stable data trends, it may obscure localized outbreak patterns.

MOH Area-Level Analysis

Dengue case data will also be analyzed across 13 Medical Officer of Health (MOH) areas within Colombo District. These areas represent the operational units of Sri Lanka's public health system.

Geospatial processing will be conducted using:

- QGIS for shapefile preparation and spatial validation

- GeoPandas (Python) for programmatic spatial joins between case data and geographic boundaries

A spatial resolution sensitivity analysis will compare machine learning model performance when trained on:

- District-level datasets
- MOH-level datasets

The evaluation metrics F1-score and RMSE will determine which spatial resolution provides the most accurate dengue prediction results.

3.9 Validation Strategy and Performance Metrics

The quality of the dataset variants will be evaluated through downstream machine learning experiments.

The same machine learning models—Random Forest and XGBoost—will be trained on each dataset variant using fixed hyperparameters. This ensures that differences in model performance arise solely from differences in the data engineering strategies.

Two types of predictive tasks will be evaluated.

Classification Task

Predicting dengue outbreak risk levels (Low, Medium, High).

Evaluation metrics:

- F1-score (macro)
- Precision
- Recall
- ROC-AUC

Regression Task

Forecasting the number of dengue cases per week.

Evaluation metrics:

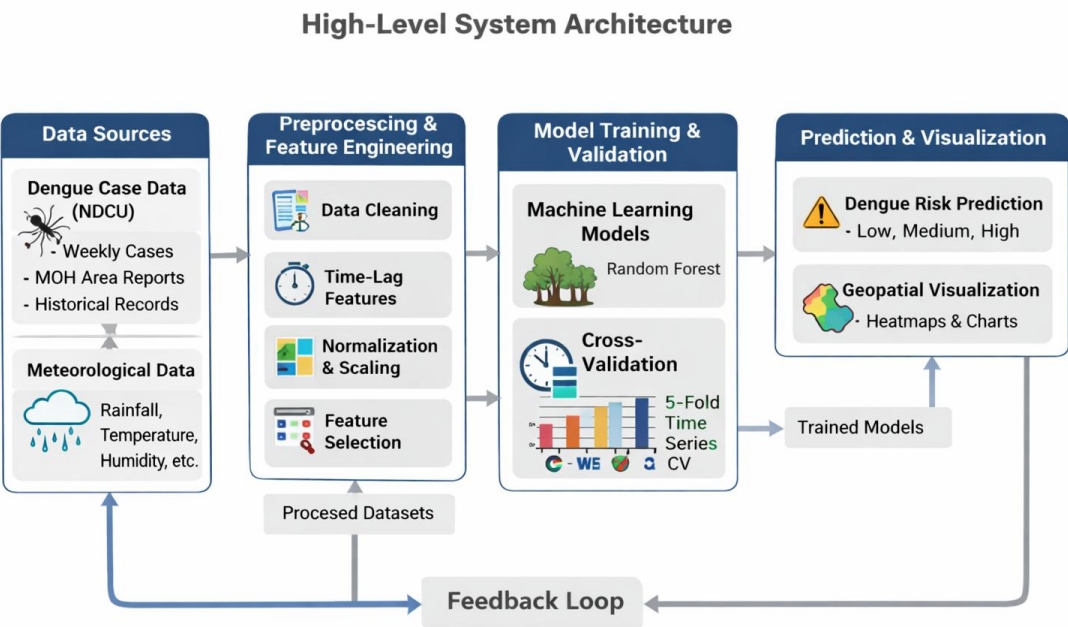
- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)

To ensure robustness, 5-fold time-series cross-validation (walk-forward validation) will be applied. This method preserves the temporal structure of the data and evaluates model stability across multiple forecasting periods.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system consists of four major components: data engineering, machine learning prediction, explainable analytics dashboards, and a knowledge-augmented NLP module. Historical dengue and weather data are first collected and preprocessed to generate machine-learning-ready datasets.

Machine learning models analyze climate patterns and generate dengue outbreak predictions. The results are visualized through interactive dashboards that display risk levels, trends, and spatial heatmaps. The NLP module retrieves dengue-related knowledge from curated data sources and generates explanatory insights using retrieval-augmented generation. Together, these components form an integrated dengue intelligence platform.



5. USER REQUIREMENTS

Stakeholders include public health authorities, researchers, and healthcare organizations.

Functional Requirements:

1. Predict dengue outbreak risk levels.
2. Visualize dengue trends and climate correlations.
3. Provide explainable insights for predictions.
4. Retrieve dengue knowledge from documents.
5. Generate automated health intelligence reports.

Non-Functional Requirements:

- High prediction accuracy
- Secure data handling
- Fast system response
- User-friendly visualization dashboards

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